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FLOW FIELD MIXING CHARACTERISTICS OF AN AERO-ENGINE COMBUSTOR - PART I: EXPERIMENTAL RESULTS

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Abstract

A model combustor with wide optical access was built for the investigation of isothermal mixing. The planar Quantitative Light Scattering Technique (QLS) was used for the measurement of the concentration of fluid injected through the dilution jets into the combustor liner. The velocity distribution in the combustor was measured with a 2-D Laser Doppler Anemometer. The derivation of the Mie scattering as a quantitative measure for concentration distributions is given, the image corrections methods are described and the experimental setup used for the investigation is discussed together with an error estimation. The errors of the measured mean concentrations can be limited to 5%. The combination of the measurement of the velocity and the concentration of jet fluid as a passive scalar qualified the experiment as suitable test case for the verification of the capability of various CFD codes to model turbulent mixing in the presence of swirling recirculation. The results show a strong coupling of the jet mixing to the structure of the primary zone flow. The variation of the jet to mainstream momentum ratio with the position of the jet relative to the burner nozzles and the pressure distribution created by the recirculation influences the jet penetration and backmixing of jet fluid in the primary zone notably.

Introduction

This is the companion paper of: Flow field mixing characteristics of an aeroengine combustor part II numerical results.[1] Both papers limit themselves to the isothermal flow of the combustor. The experimental work was carried out as a part of a cooperative program between the engine manufacturer and the research establishment, where optical techniques were used to investigate the mixing and reaction in a combustor sector. Since improving the homogenization of the mixture in the combustor was an important part of the solution strategy to obtain a combustor with reduced emissions, there was a strong interest to gain direct access to the quantities controlling the mixing and verify the predictive quality of the computational tools. A useful preliminary step in the verification process is the separation of the purely fluid mechanical mixing before considering the added complexity of the influence of reaction and density changes.

Therefore an optical model of the combustor for isothermal flows was built by MTU and it's mixture and flowfield measured by DLR. For the mixture of the air flows entering the combustor, the method of quantitative light scattering (QLS) of Mie scattering particles was enhanced and the flow field was measured by LDA. The problems of obtaining a quantitative measure of the mixture of multiple streams of air are discussed and the technical solutions are described. Measurements of the flow field are presented with a focus on the jet mixing into a swirling primary zone flow, that was important for the rich-quench-lean combustion to be realized in the combustor.

Model Combustor

The model combustor for investigations of the combustor characteristics under isothermal conditions at ambient pressure was a scaled down planar sector element of the annular aero-engine combustor developed by MTU AeroEngines within the Engine3E project. The scaling was necessary in order to fit a sector combustor with a minimum of three segments, each comprising one burner, into an existing test facility with an open test section of 180 mm. The underlying principle of the scaling was to keep the mass distribution (main/secondary), the pressure drop and the momentum ratios between the dilution jets

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and the combustor main flow the same as on the original annular Engine3E combustor.

Fig. 1 shows the model combustor assembly in three different views. The combustor casing, the separation walls between the main and secondary flow as well as the combustor liner were made from aluminum. The combustor top wall as well as the combustor top and side walls were partially carried out as windows in order to enable the optical access for the optical measuring techniques. Screens were installed both in the main and the secondary channels in order to straighten the flows before entering the combustor. In Fig. 2, a photograph shows the model combustor together with the LDA equipment.

The combustor consisted of a primary zone with constant cross section and a convergent dilution zone. At the combustor head primary air was introduced by means of three airblast atomizer nozzles and through slots at the top and bottom end of the heat shield representing the combustor cooling air. The dilution air was injected into the main flow at the transition from the primary zone to the dilution zone through three rows of dilution holes which were arranged in a staggered configuration in order to achieve a rapid mixing with the main flow.

LDA-Measurements

The velocity measurements were performed under isothermal and atmospheric conditions using a modular two-component DANTEC laser-Doppler anemometer. The velocity components were measured with the green (λ=514.5 nm) and blue (λ=488 nm) light of an argon laser driven with less than 0.5W total light power.

The intersection angles of the two couples of light-beams were seven degrees. The signal light was received under near forward-scatter to obtain a high signal rate together with good signal quality.

The solid scattering particles consisted of SiO_2. They were nearly ball-shaped and fairly monodisperse with diameters of 0.8 micrometers with a size distribution width of 20% of the mean value. The powder was dispersed into a low bypass air flow which was introduced into the hoses for the main air flow upstream of the model combustor.

Two DANTEC BSA's were available for signal acquisition and analysis. A computer-controlled three-axis traversing system scanned the flow field by moving the optics in a horizontal plane and the model combustion chamber vertically.

All measurements were made without any coincident condition of the BSA's . Thus we were able to realise the necessary high signal rates especially for an FFT-analysis of the velocity fluctuations along a frequency range of at least some KHz.

The measuring light beams were introduced into the model chamber through one of the end plates of the chamber and the scattered signal light was received from the opposite direction. In the case of the measurements near the nozzle exit the optical system was inclined against the chamber base plate to prevent the blocking of the axial-velocity measurement beam.

The reported measurements refer to the flow field of the central sector only. The other sectors served to simulate the flow interactions of a multi-sector. The measurements were additionally extended to the near-wall region downstream in order to study the flow field around the dilution jets.

Mixing fraction measurements

Many of the existing techniques for mixing fraction measurement are based on point wise measurements, like chemical or thermal probe sampling.[2] It is obvious that probe sampling underlies severe drawbacks in combustion research, since the method is time consuming and the probe influences the mixing process where it is most intense: in the high gradient regions. Optical techniques are promising, since they have the potential to give planar information about the mixing process without disturbing the flow field.

Light sheet techniques based on Mie-scattering or laser induced fluorescence (LIF) have been demonstrated to be basically suited for planar mixing fraction measurements of isothermal flows. Being dependent on high power lasers to overcome the comparatively low-signal level, the LIF-technique has not achieved an importance comparable to the more efficient Mie-scattering technique in the past. Today LIF becomes increasingly attractive due to the progressing development of lasers. The first system was set up by Long et al. [3] and later employed by several other researchers.[4,5] In spite of the advantages of high signal level and ease of optical set up, the Mie-scattering technique requires quite a lot of effort in data processing to provide accurate, quantitative data. Although the aforementioned authors have already introduced image-processing procedures to correct for background, sensitivity vignetting effects, etc. and for the light sheet intensity distribution, they did not take into account extinction and scattering direction effects, which are very significant in practice. The Mie-scattering system introduced in this paper accounts for these additional effects thus improving measurement quality, making the method a quantitative technique.

Measurement Principle of the Quantiative Light Sheet (QLS) Technique

The QLS-technique described in this chapter was earlier mentioned by P. Voigt and R. Schodl [6], P. Voigt [7] and C. Hassa et al. [8]. The light sheet method uses scattering of light by small particles marking the flow. The inflow must consist of at least two separate streams, allowing to inject tracer particles to only one of them: the other streams must remain unseeded. The mixing of the seeded injection flow (index 1) with the unseeded main flow (index 2) is analyzed by illuminating the interaction region with a laser light sheet and recording the scattered light from this region on a camera film or CCD-chip. Under the test condition, which will be specified in the following, the gray values of the detected image correspond to the main concentration of the seeded flow in the mixture.

The imaging of the light sheet onto the CCD camera chip is shown schematically in Fig. 3. Depending on the imaging scale, a single pixel on the chip with the area $\Delta x_{im} \bullet \Delta y_{im}$ corresponds to a certain area element $\Delta x \bullet \Delta y$ in the light sheet at position (x,y). The thickness Δz of the light sheet then determines a volume element which contains a certain number $N_p(x,y)$ of particles. As long as the flow conditions are in steady state and seeding is constant, this number N_p is constant in time and independent of the flow velocity, since with increased velocity more particles per unit time enter the volume element but also leave it. It depends only on particle concentration C_p and local fluid density $\rho_f(x,y)$. The grey level $g(x,y)$ indicated by each pixel is proportional to the scattering power $P_{sca}(x,y)$ collected by the camera lens from all particles in the volume element during the exposure time Δt. In the simplified case of uniform particle distribution, the grey level, which represents the collected light energy, is:

$$g(x,y) \sim \Delta t \cdot P_{sca}(x,y) N_{P(x,y)} \ . \qquad (1)$$

Considering a mixing of two flow components the mass concentration of component one C_1 is determined by

$$C_1(x,y) = \left(\frac{\dot{m}_1}{\dot{m}_1 + \dot{m}_2} \right)_{x,y} \ . \qquad (2)$$

This value is equivalent to the ratio of the total mass portions within the volume element,

$$C_1(x,y) = \left(\frac{M_1}{M_1 + M_2} \right)_{x,y} \ . \qquad (3)$$

Since flow component 2 is unseeded, the total number $N_p(x,y)$ of particles in the volume element corresponds only to the portion $M_1(x,y)$ of the mass, so

$$N_p(x,y) = C_{p1} \cdot M_1(x,y) \ . \qquad (4)$$

C_{p1} is the particle concentration of flow component 1 which is supposed to be constant. The particle size distribution should also not vary when mixing with the unseeded air:

$$C_{P1} \neq C_{P1}(x.y)$$

With equation (3) and (4) and the mass contained in the volume element

$$(M_1 + M_2)_{x,y} = \rho_f(x,y) \cdot \Delta x \cdot \Delta y \, \Delta z \qquad (5)$$

the relation (1) can be transformed into

$$g(x,y) \sim \Delta t \cdot P_{sca}(x,y) \cdot C_{P1} \cdot \rho_f(x,y) \cdot \Delta x \cdot \Delta y \cdot \Delta z(y) \cdot C_1(x,y) \qquad (6)$$

The collected scattered light from a single particle is given by

$$P_{sca}(x,y) \sim C_{sca}(d_p, x) \frac{P_L(x,y)}{\Delta y \cdot \Delta z(y)} \qquad (7)$$

with $C_{sca}(d_p,x)$ the scattering cross section as a function of particle diameter and scattering direction which varies predominantly with the x-coordinate and the local laser power $P_L(x,y)$. In general the scattering cross section depends also to a small extend on the y-coordinate because of polarization effects. When non polarized or circularly polarized laser light is used this dependence can be neglected.

The grey level proportionality is then given by

$$g(x,y) \sim \Delta t \cdot C_{sca}(d_p, x) \cdot P_L(x,y) \cdot C_{p1} \cdot \rho(x,y) \cdot \Delta x \cdot C_1(x,y) \qquad (8)$$

It is remarkable that the grey level is independent of the light sheet thickness Δz. With increasing Δz the volume of the element and the number of particles therein increase but the light intensity decreases by the same ratio. Since it is practically impossible to calibrate the grey level g against mass concentration C_1 because of too many parameters influencing the light detection, the QLS technique has been set up as a relative method, i.e. all grey values in the detected image are related to the value at the location (x_0,y_0) of the injected jets, where the seeded particle flow penetrates the cross flow. This point (index 0) is used as a reference point for 100 percent jet flow concentration. The grey value ratio hence results to

$$\frac{g(x,y)}{g(x_0,y_0)} = \frac{C_{sca}(d_p,x)}{C_{sca}(d_p,x_0)} \frac{P_L(x,y)}{P_L(x_0,y_0)} \frac{\rho_f(x,y)}{\rho_f(x_0,y_0)} \frac{C_1(x,y)}{C_1(x_0,y_0)}. \tag{9}$$

Since both grey values are taken form the same image, the exposure time can be eliminated.

Because of the constant particle concentration of flow component 1 and the unvarying size distribution, the scattering correction is spatially only dependent on the x-coordinate

$$\frac{C_{sca}(d_p,x)}{C_{sca}(d_p,x_0)} = \frac{C_{sca}(x)}{C_{sca}(x_0)} \tag{10}$$

and therefore equation 9 can be reduced to

$$\frac{g(x,y)}{g(x_0,y_0)} = \frac{C_{sca}(x)}{C_{sca}(x_0)} \frac{P_L(x,y)}{P_L(x_0,y_0)} \cdot \frac{\rho_f(x,y)}{\rho_f(x_0,y_0)} C_1(x,y) \tag{11}$$

The concentration $C_1(x_0,y_0)$ in the core of the seeded jet flow is set to 1.0 corresponding to 100% of component 1.

We assume that the laser power distribution in the light sheet as well as the directional dependence of the scattering cross section is known. Then the grey level ratio is a measure of mass concentration only, provided that either the density field is known or constant in the measurement region. The latter case is considered for QLS applications and equation 11 becomes

$$C_1(x,y) = \frac{C_{sca}(x_0)}{C_{sca}(x)} \cdot \frac{P_L(x_0,y_0)}{P_L(x,y)} \cdot \frac{g(x,y)}{g(x_0,y_0)} \tag{12}$$

Hence the practical application of the QLS-technique is restricted to isothermal mixing processes under constant pressure conditions of flow components of the same fluid.

The influence of the scattering direction is significant and requires corrections. In Fig. 4 a typical QLS setup is shown with a CCD camera observing a certain region *G* of a light sheet and detecting the scattered light particles. With different positions *x* in the region *G* the scattering angle ϑ between incident laser beam and observation direction varies, thus on the right corner of region *G* low intensity back scattered light and on the left corner high intensity forward scattered light is observed. This effect which increases with increasing viewing angle is show in Fig. 4, where the scattering intensity $I(\vartheta)$ is plotted as a function of ϑ over the entire range of the region *G*. This measurement can be made either at the actual experimental set up if homogeneous seeding of the flow in the measurement region is possible or in a separate experimental set up where care must be taken to keep the general arrangement and the seeding properties identical. The result is the distribution $C_{sca}(x)$ which is needed in equation (12) for corrections.

The Laser power distribution in the laser sheet is the second parameter which must be known. It is mainly determined by the light sheet generation. A constant intensity distribution over the light sheet height (*y*-coordinate) is very important for achieving best accuracy with a given dynamic range of the camera sensitivity. Sweeping a laser beam periodically over the light sheet height with a rotating polygon mirror is a rather efficient solution to closely approach the desired constant intensity distribution. In this case a CW-laser, as for instance an Ar^+-Laser, must be used as a light source. The exposure time should be significantly longer than the sweep time or an exact integer multiple of it. The light sheet generating optics should be designed such that the laser beam direction remains approximately constant while the beam is moved up and down. In this way a parallel light sheet is formed with constant intensity along the propagation direction *x*. The intensity distribution along the light sheet height can be measured. With the known value

$$P_L(x,y) = P_L(y) \tag{13}$$

equation (12) is finally determined.

However this conclusion can generally not be drawn because there are two physical phenomena related to the particle laser light interaction, which have not been considered yet. They influence the laser power distribution as well as the scattering characteristics in the measurement region.

Measurement Errors

One effect is extinction, which is the attenuation of a propagating light beam by the scattered light emitted from particles. As shown by P. Voigt[7] one can correct for this intensity loss along the x-coordinate when the laser power distribution $P_L(y)$ is known at the entrance and exit of the measurement region and when extinction is the only existing effect.

However in the case of light scattering on particles there are in parallel effects of multiple scattering overlapping the extinction, rendering these corrections impossible. Multiple scattering is a secondary effect in scattering experiments. It denotes the fact that scattering particles can be light sources for adjacent particles. This effect depends on the particle load. The existence of secondary scattering has been proven in a number of experiments and different aspects have been simulated numerically. An example of multiple scattering effects is shown in Fig. 6 by the virtual thickening of a laser beam: A laser beam passes through a chamber with homogeneous seeding. The laser beam enters from the left side. On the outlet side the transmission ratio τ is measured by a photodiode. The chamber is seeded with four different particle loads and images have been taken. In the upper left image no extinction could be measured. The beam looks thin. For the following experiments the seeding was intensified successively. As a result the laser beam in the image increases it's thickness accordingly. But this expansion is not constant along the beam. In the first quarter of the path the expansion increases, due to the strong secondary scattering in the forward direction. After reaching the maximum diameter, the expansion decreases, caused by the coexisting extinction effect on the centerline.

The correction of multiple scattering is the most challenging part of the post processing, since the effects are fully three dimensional. A number of different approaches have been tested already, but none of them with sufficient results yet. Since extinction and multiple scattering are simultaneous effects, correction algorithms are even more complicated and not available yet. A full correction being impossible, there is only one solution: to keep these errors small. For that reason the seeding must be adjusted to the lowest possible level which can be accepted with regard to the camera sensitivity. In practice the seeding level should be kept so small that extinction along the light sheet path does not exceed 10%. Predicted by a detailed error analysis given in Voigt[9] and proved by comparative tests, the remaining measurement error does not exceed 5% and reaches this maximum value where multiple scattering effects are most important in regions of steep mixing fraction gradients.

Table 1 shows the influence on measurement accuracy of the most important error sources before and after correction for mixing fraction mean values.[8]

Experimental Set-Up

The experimental set up consists of an Ar^+-laser (1W power) fiber linked to a light sheet generating box in which a polygon scanner with 20 facets is rotating with controllable speed. The scanning laser beam generates the light sheet propagating with constant height (max. height ≈ 120 mm). A Peltier-cooled CCD camera with a dynamic range of 12 bit takes the picture viewing normally to the light sheet. When light reflection on model surfaces cannot be avoided, these were painted with fluorescent dye and a laser line filter was inserted to the camera objective to reduce these reflections to a great extent.

The measurements start with a background image taken with the conditions: laser on and seeding off. From the signal image taken with seeding on in one flow component the background image is subtracted and different correction procedures for pixel sensitivity, image distortion and scaling, scattering direction and laser power distribution and if possible extinction are applied. After that, the reference value $g(x_0,y_0)$ indicating the 100% level in the in the image is determined and the corrected mass concentration map is established. Further application of various graphical procedures deliver the final mass concentration contour plot.

Being restricted to low seeding levels, even a highly sensitive cooled CCD camera requires several seconds of exposure time to achieve results of good quality supposing an Ar^+-laser is used as light source. That limits this set up to mean value measurements. Instantaneous measurements with much shorter exposure time (~70µs) have also been carried out but require an intensified CCD camera and are less accurate.[10]

Results

Velocity

The position of the LDA optics as depicted in Fig. 2 allowed to obtain measurements of the axial and radial velocity. Measurements were performed in the center plane and in the plane of the axis of the first jet rows in the near field of the jets. The measurement grid was 2 mm in the axial direction up to x = 15 mm and from then on 5 mm and 0.5 mm in the y and z direction.

In Fig. 7 the mean axial velocity U and radial velocity V are depicted as arrows, the U velocity is additionally reproduced by the false color scheme. The contour of the liner shows the position of the nozzle, which is not in the middle of the combustor height, as well as the position of the holes of the circumferentially staggered mixing jets.

The typical features of an aeroengine combustor flowfield can be discerned: The expanding, swirling nozzle flow with an inner and outer recirculation zone. The inner recirculation zone is closed by the jets from the first row of dilution holes. However the

mixing zone flow is rather peculiar as the three axially and tangentially staggered rows of jets produce large three-dimensional flow patterns, which should lead to the desired quick mixing of the dilution air and thus to improved homogenization. The high penetration of the jets into the opposite half of the combustor volume is a desired feature for tangentially staggered jet rows,[2] but would be less in the combusting case, as the jet to mainstream momentum ratio decreases with the acceleration of the main flow connected to the combustion in the primary zone. To some extent, this can be simulated by changing the repartition of mass flows and thereby velocities according to the zonal temperature levels and this has been done for the QLS measurements, but it will not be presented here, as the aim of this contribution is the generation of validation data for the CFD calculations of Part II.

A closer look on the nozzle near field of Fig. 8 reveals a recirculation, that extends upstream across the first measurement location at x = 1mm. The swirling flow expands with a moderate rate up to x = 14 mm. Here the recirculation zone widens quickly under the influence of the backpressure that is exerted by the perpendicular jets of the mixing zone. The corresponding picture for the circumferential velocities in Fig. 8 has a zero velocity contour which is deflected due to the influence of the three dimensional flow pattern of the jets producing an asymmetric recirculation zone. The upstream movement of jet fluid in the upper half of the combustor suppresses the toroidal recirculation zone that is present for symmetric jet configurations, as in this plane only the lower nozzle flow recirculates, whereas the upper part is deflected by the first jet and it is mostly jet fluid filling up the upper half of the recirculation. The asymmetric position of the nozzle slightly above the symmetry line of the liner also influences the penetration of the jets, as the momentum ratio on the downside is slightly higher, leading to a lower jet angle, higher negative axial velocities of the first jet in the upper half and finally a jet angle of the second jet toward the primary zone, which is even more negative than the angle of the liner wall. The variation of the momentum ratio of the jets along the lateral direction depending on the distance to the nozzle produces a variation in jet angles in the axial and to a lesser extend in the lateral direction. The resulting spatial distribution of jet cores tends to widen the effective jet spacing and thus the intensity of shear and mixing and might inhibit the desired jet penetration. The effect could in theory be corrected or even be used to enhance mixing by creating counterswirling flows when the jet diameters are varied accordingly in the lateral direction, as has been suggested by[11].

As the lateral velocity component was not measured, a turbulent quantity $U_{turb} = (u'^2 + v'^2)^{0.5}/U$ was constructed, that is represented in a false color scheme in Fig. 9 together with the mean velocity vectors. Not surprisingly, comparison with Fig. 7 shows the highest intensities at the boundaries of the inner and outer recirculation as well as in the region between the first and second and second and third jets. Since the measured values are very high, note the logarithmic scale on Fig. 9, these regions are candidates for coherent large scale fluctuations or even periodic oscillation, and a periodic instability was actually found for the used burner nozzle,[12] however for a systematic investigation it would be necessary to make simultaneous or phase averaged 2-D velocity measurements, which is beyond the scope of this contribution.

Concentration Measurements

The concentration measurements were made in 61 axial-radial planes with a separation of 1 mm covering the middle sector of the model combustor. Because of the convergent part of the combustor, 3 light sheet directions were necessary to cover the full cross section of the combustor with light sheets entering from the combustor exit: 1 following the upper liner contour, one following the lower liner contour and one parallel to the burner axis. The resolution in the axial and radial direction is about 0.25 mm, however for the computational ease of the post processing 4*4 elements were integrated to give a 1 mm^2 point in the plots.

Fig. 10 shows the mean concentration field of the jet fluid in the central plane of the combustor. Comparison of the vector field of Fig. 7 gives a good impression of the backmixing of jet fluid in the primary zone, which is partly due to the deflection of the mean streamlines of the first jet and partly to the turbulent diffusion of jet fluid into the recirculation in the high shear regions at the lower part of the jet. In the dilution zone, the large scale jet mixing works quite well, as can be seen by the rapid homogenization downstream of the third jet. At x = 100 mm, only 50 mm downstream from the first jet, the bandwidth of the concentration of mixture has decreased to 15%. The threedimensional character of the jet mixing is depicted in Fig. 11 which shows a contour of the jet fluid concentration in the middle sector where C is $0.5<C<0.55$. The contourplot of C at z = 0 is included. Differences in the contour of the 2 jets from the second row in the positive y and z direction can be seen. They correspond to the variation of jet penetration according to the local variation of momentum flux.

Fig. 12 shows the RMS values of the concentration as computed from 10.000 pictures produced with a single sweep of the scanning laser beam at 70 µs per sweep. This representation exposes very nicely the potential core region of the first jet. Regions of high concentration fluctuation are connected with the wake of the first jet and the lower and upper part of the

backmixing region, where jet fluid is transported downstream again in an rather discontinuous fashion, as shown by inspection of single shot images.

Conclusions

The present contribution summarized the results of an experimental effort, which was directed towards a realistic and complete characterization of the steady state isothermal flow and mixing in a model combustor geometry exhibiting the complexity of modern aeroengine combustors. The Mie-scattering technique was improved to yield quantitative results for the concentration of dilution jet fluid in the combustor within a 5% error margin. This accuracy and the superior spatial resolution of the planar technique make it fit to be used for a comparative judgement on the ability of various CFD codes to predict turbulent mixing. However the experimental expenditure in terms of the required expertise and care is also rather high. Comparison of experimental and numerical results shows the limits of a purely steady state approach. Capturing the non steady state and periodic effects and weighting their relative importance for the description of the macroscopic features of the combustion will be a task for the years to come.

References

[1] Noll, B., Kessler, R., Theisen, P., Schütz, H., Aigner, M., Flow field mixing characteristics of an aero-engine combustor – part II: Numerical simulations, AIAA 2002-3708, 2002

[2] Migueis, C. E. S. S., Untersuchungen zur Optimierung der Mischzone einer fett-mager gestuften Ringbrennkammer , DLR-FB--96-33, ISSN 0939-2963, 1996

[3] Long, M. B., Chu, B. T., Chang, R. K., Instantaneous Two-Dimensional Gas Concentration Measurement by Light Scattering, AIAA Journal, Vol. 19, No. 9, Sept. 1981

[4] Vranos, A., Liscinsky, D. S., Planar Imaging of Jet Mixing in Cross Flow, AIAA Journal, Vol. 26, No. 11, Nov. 1988

[5] Vranos, A., Liscinsky, D. S., True, B., Holdeman, J. D., Experimental Study of Cross-Stream Mixing in a Cylindrical Duct, NASA Technical Memorandum 105180, AIAA-91-2459, 1991

[6] Voigt, P., Schodl R., Using the Laser Light Sheet Technique in Combustion Research, Paper 48, AGARD CPP 598: Advanced Non-Intrusive Instrumentation for Propulsion Engines, 1998

[7] Voigt P., Non-linear effects in planar scattering techniques: proof of existence, simulation and numerical corrections of extinction and multiple scattering, 9th Int. Symp. on Appl. of Laser Tech. to Fluid Mechanics, Lisbon, Portugal, 1998

[8] Hassa, C., Migueis, C. E. S. S., Voigt, P., Design Principles for the Quench Zone of Rich-Quench-Lean Combustors, Paper 18, RTO-MP-8: Design Principles and Methods for Aircraft Gas Turbines Engines, 1999

[9] Voigt, P., Entwicklung und Einsatz eines Laserlichtschnittverfahrens zur quantitativen Konzentrationsmessung bei Mischungsprozessen DLR-FB--1999-41, ISSN 1434-8454, 1999

[10] Voigt, P., Theisen, P., Experimental analysis of unsteady mixing in a RQL combustor segment aimed for validation of PDF models in the TRUST code, Notes on Numerical Fluid Mechanics, Vol. 72, Vieweg, 1999

[11] Blomeyer, M., Entwicklung von Auslegungskriteriien für die Mischzone einer luftgestuften Gasturbinen Ringbrennkammer, Fortschrittberichte VDI, Reihe 6 Energietechnik, Nr. 427, ISSN 0178-9414, 1999

[12] Schütz, H., Noll, B., Lehmann, B., Aigner, M., Numerical simulation of vortex shedding at the exit of an air/fuel nozzle, Eurotherm 67, ECOS-2000 Conference, pp 2143-2154, Twente, The Netherlands, 2000

Error source	Error (in percent)		Method of correction
	Before Correction	After Correction	
Background / reflections	20 %	< 1%	Subtraction of background image suppression of reflections by painting
Pixel sensitivity	9%	<1%	Division by sensitivity image
Light sheet intensity distribution	25%	<2%	Division by reference profile
Scattering directions	Up to 50% depending on image size	<2%	Division by calibration function
Extinction	<10%	<2%	Correction by shooting method
Multiple scattering	Max 5%	No correction 5%	Not available

Table 1: main error sources and their influence on mixing fraction mean value

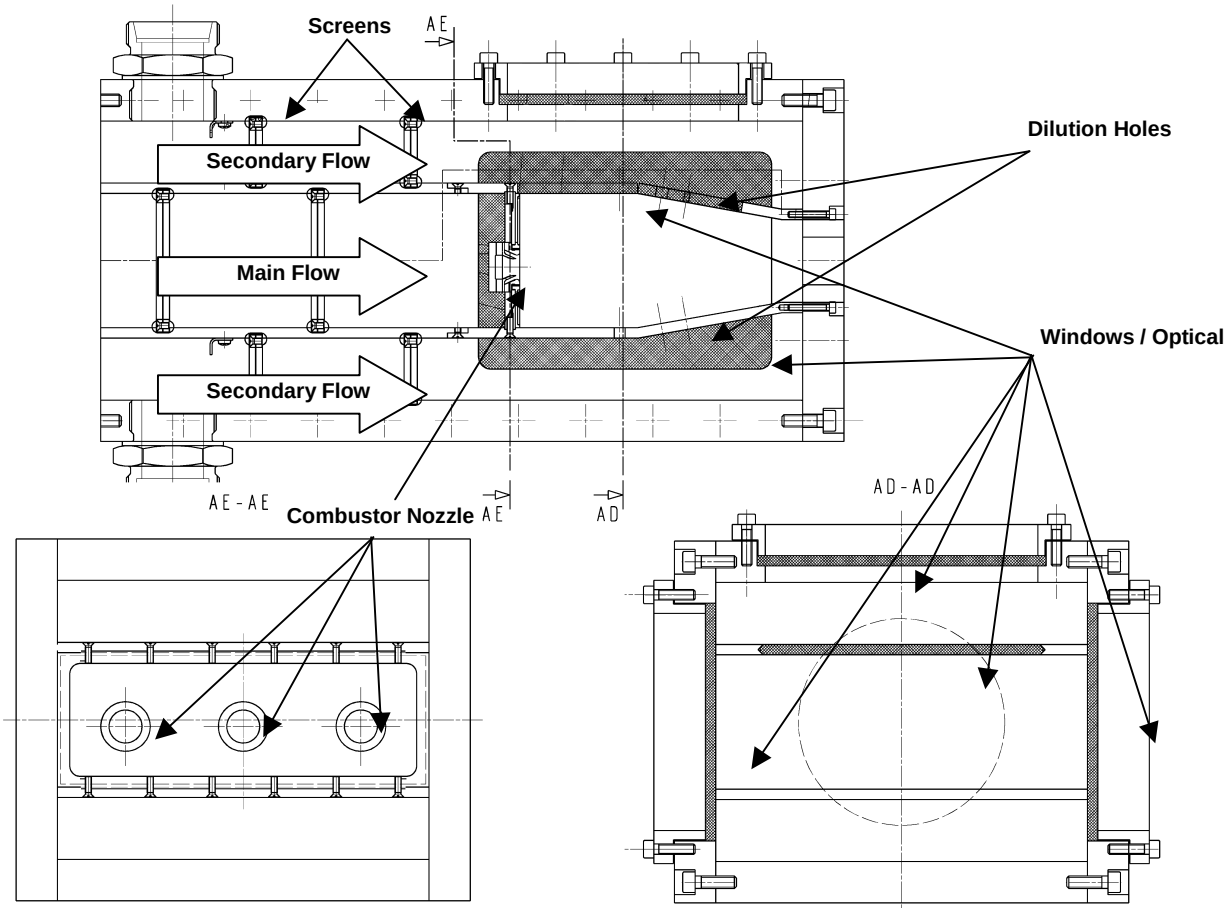

Fig 1: Sector Combustor Assembly

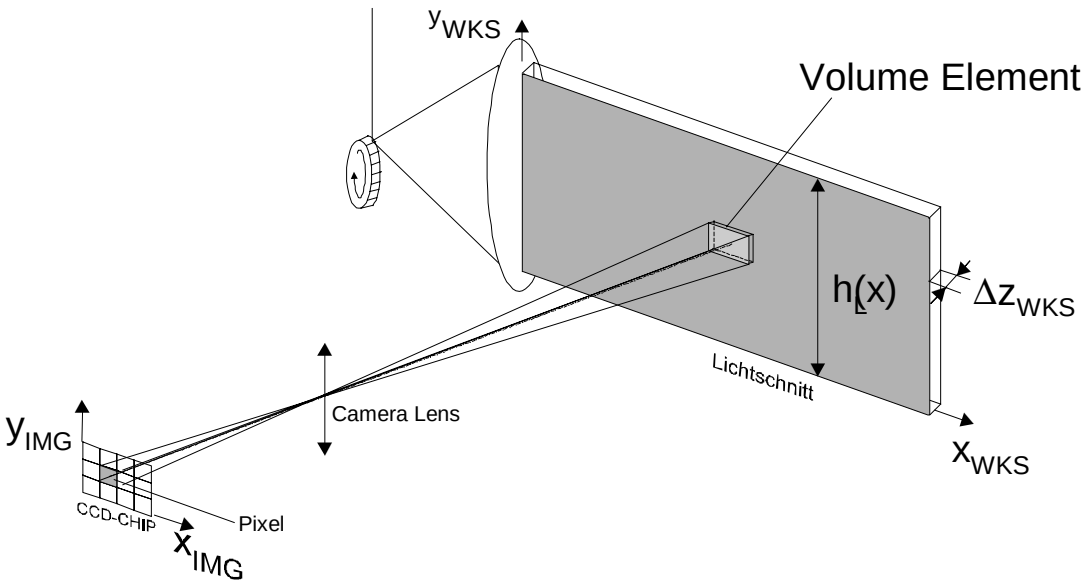

Fig 3: Imaging the light sheet onto the CCD camera chip

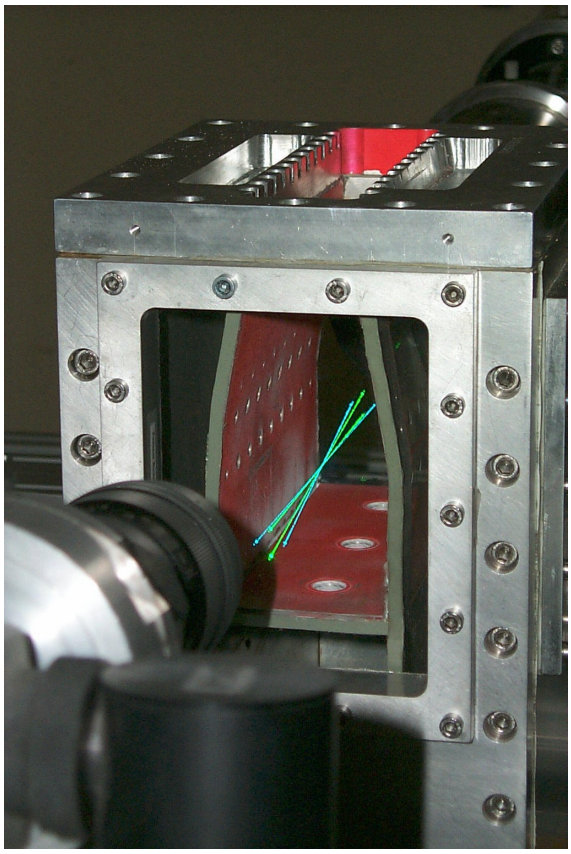

Fig. 2: Photograph of LDA setup

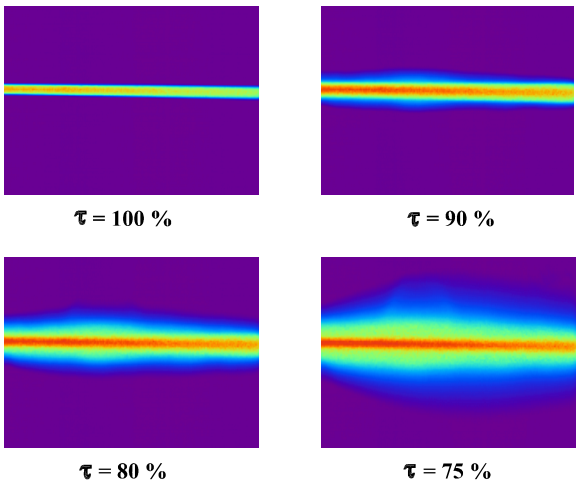

Fig. 6: Multiple scattering corona around a stationary laser beam

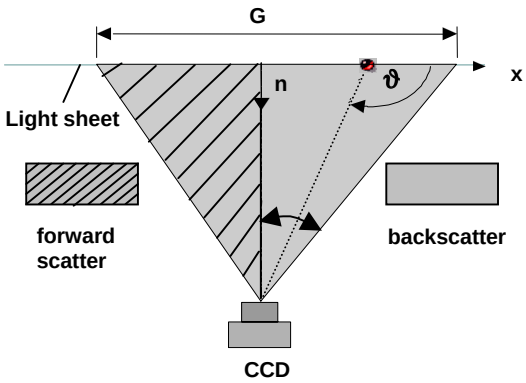

Fig. 4: Influence of scattering direction ((ϑ = ϑ(x))

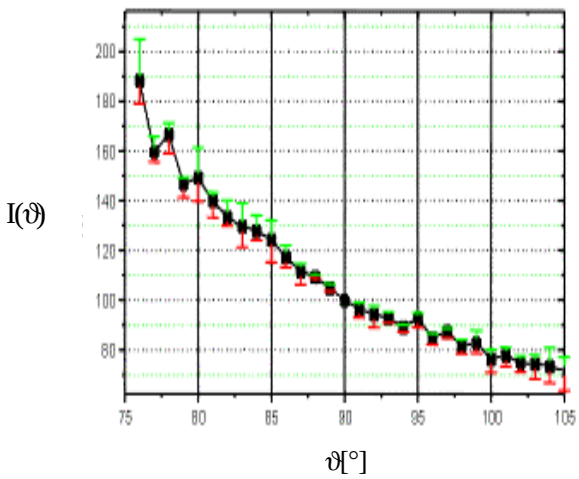

Fig. 5: Scattering intensity as a function of scattering direction ϑ.

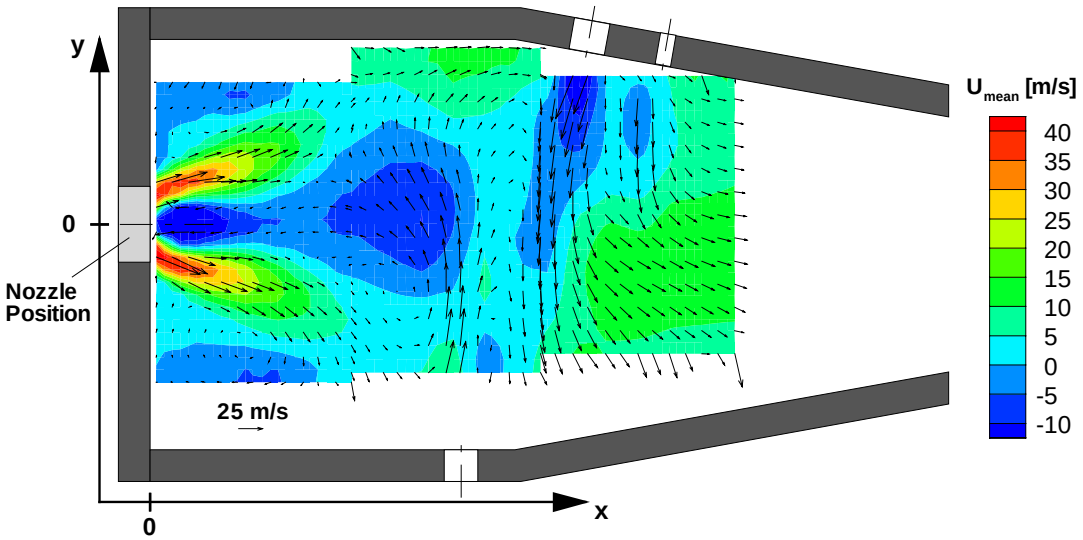

Fig. 7: Mean velocities in axial and radial direction at the combustor center plane (z = 0 mm) measured by LDA

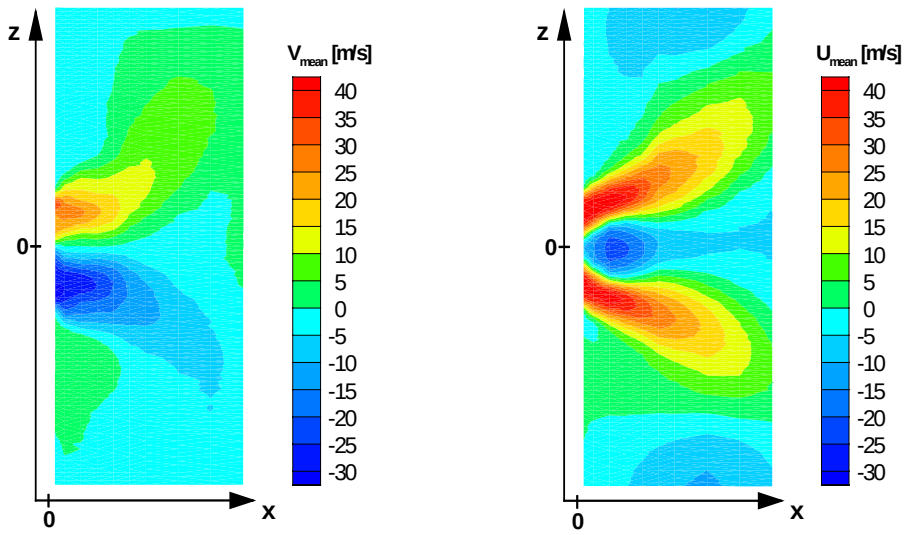

Fig. 8: Velocities in the nozzle near field at the combustor center plane (y = 0 mm) measured by LDA; left figure: lateral velocity, right figure: axial velocity

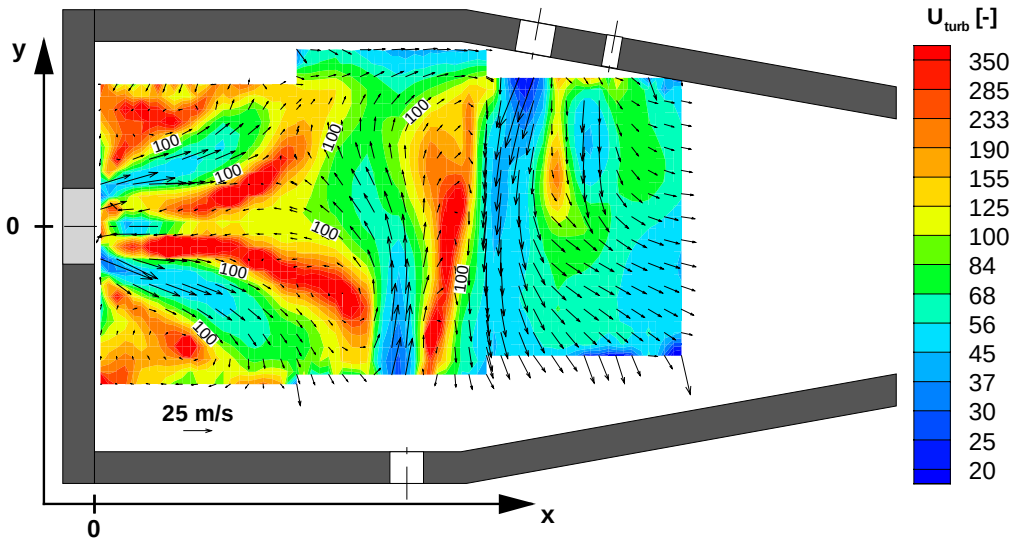

Fig. 9: Turbulent distribution and mean velocities at the combustor center plane (z = 0 mm) measured by LDA

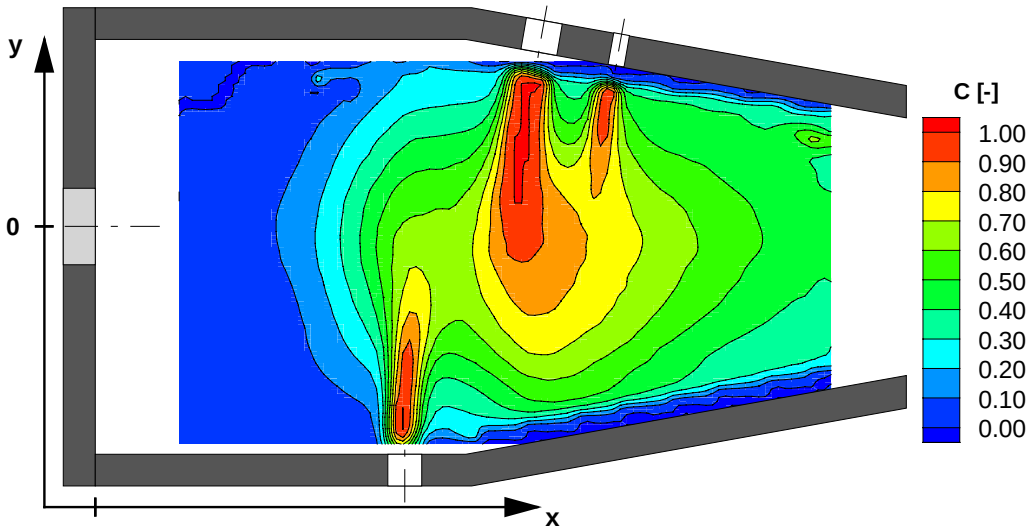

Fig. 10: Mean concentration field C of the jet fluid at the combustor center plane (z = 0 mm) measured by QLS

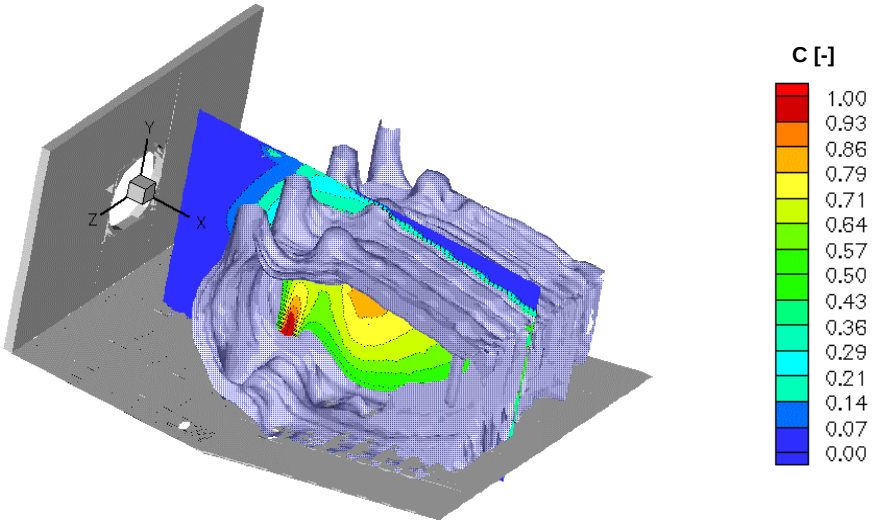

Fig. 11: 3D mixing field at the middle sector of the combustor and mean concentration field C in the combustor center plane (see Fig. 10). The iso surfaces represent the regions where the concentration of the jet fluid is located between 50% and 55%.

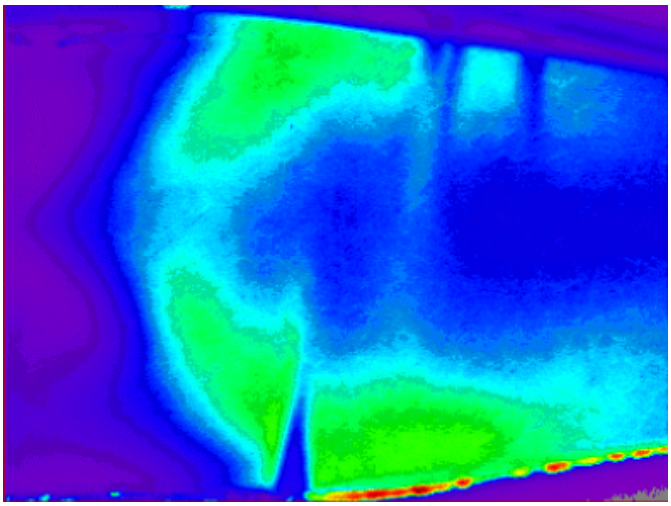

Fig. 12: RMS values of the concentration of the jet fluid at the combustor center plane computed from 10.000 pictures measured by QLS (green: high RMS, blue: low RMS)